Skooly Domain Model

This document describes the domain model for the Skooly school management platform. Entities marked with *(planned)* are described in the system design but not yet in the schema.

Relationship Types (UML)

Symbol	Type	Meaning
◆ / *——	**Composition**	Strong ownership; child cannot exist without parent; lifecycle bound
◇ / o——	**Aggregation**	Weak ownership; child can exist independently; "has-a" collection
——>	**Association**	Uses/links to; no ownership; bidirectional or directional
<|——	**Inheritance**	"Is-a" relationship; specialization
- - ->	**Dependency**	Weak usage; one depends on another for operation

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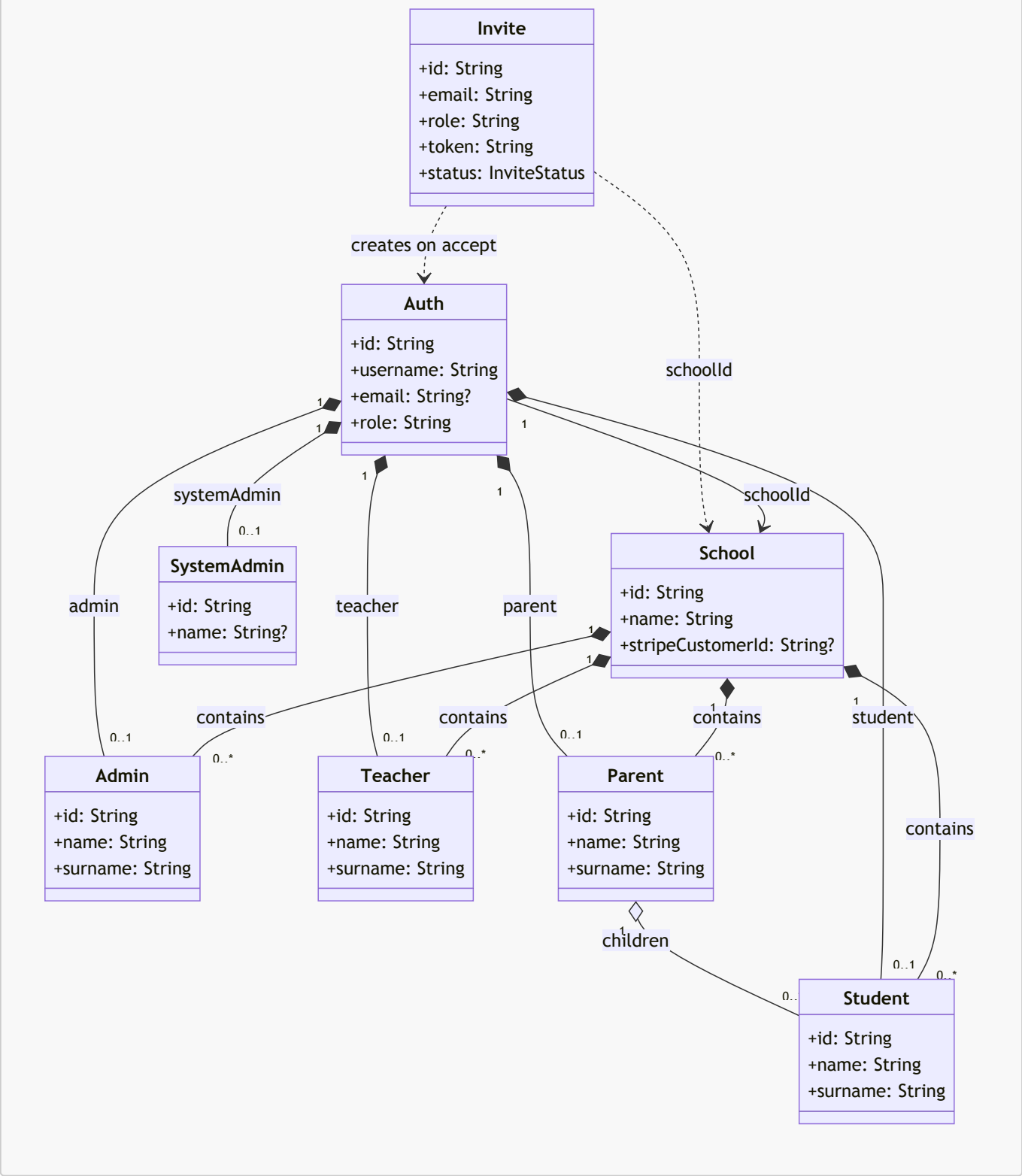
Diagram 1: Core Domain (People & Tenant)

Composition: School *owns* Admin, Teacher, Student, Parent, Auth (school-scoped). They cannot exist without a School.

Composition: Auth *owns* exactly one profile (Admin, Teacher, Student, or Parent). The profile is the identity extension.

Association: Parent ↔ Student (many-to-many via family relationship; in schema it's Parent has many Students).

Dependency: Invite (planned) depends on School and creates Auth + profile on acceptance.



Relationship summary (Diagram 1):

From	To	Type	Cardinality
School	Admin, Teacher, Student, Parent	Composition	1 : 0..*
Auth	Admin, Teacher, Student, Parent, SystemAdmin	Composition	1 : 0..1 (exclusive)
Auth	School	Association	N : 1
Parent	Student	Aggregation	1 : 0..*

From	To	Type	Cardinality
Invite	School	Dependency	N : 1

Diagram 2: Academic Structure

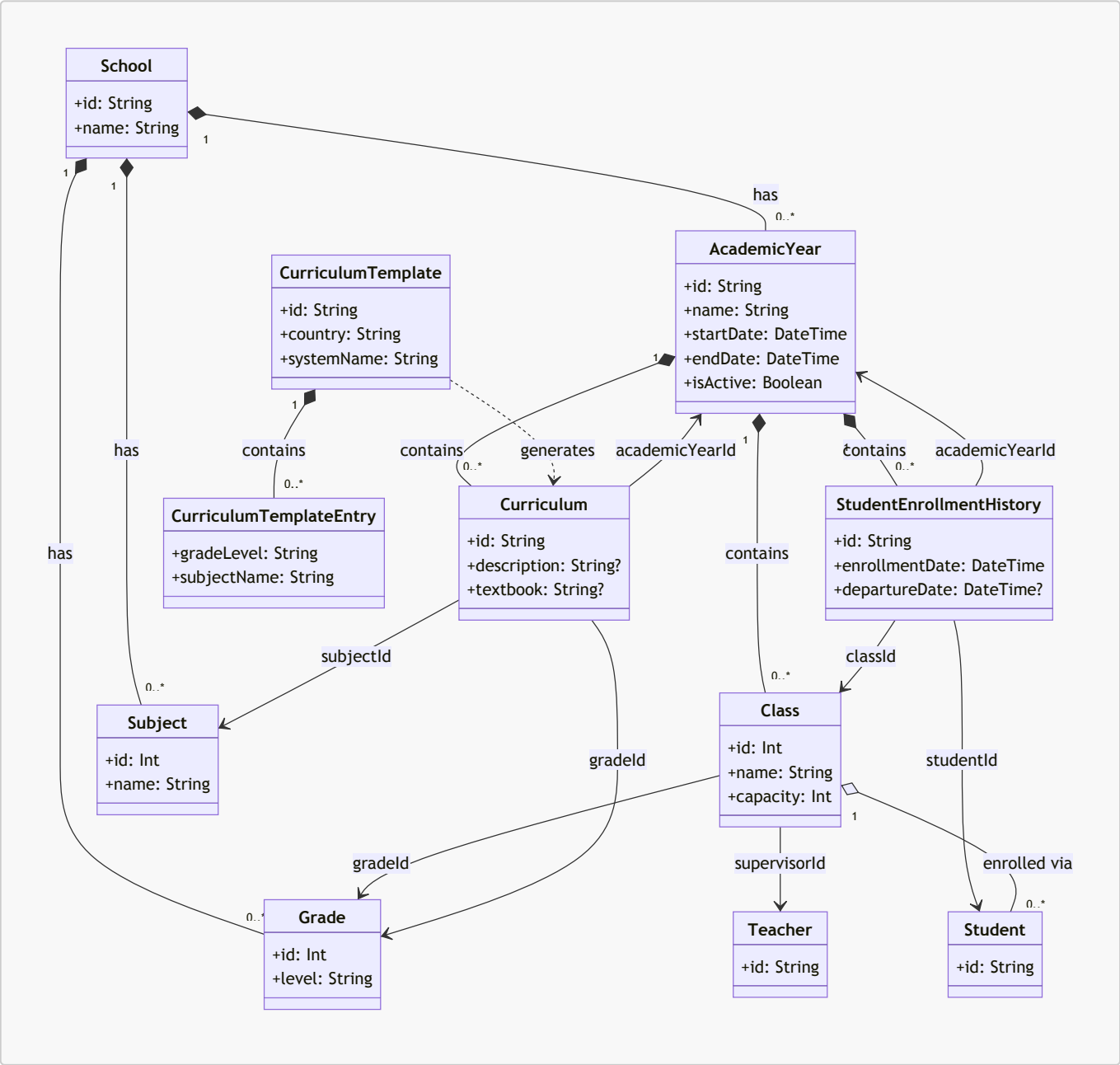
Composition: AcademicYear *owns* Class, Curriculum, StudentEnrollmentHistory. These are scoped to a year and do not exist without it.

Composition: School *owns* Grade, Subject (school-level, reusable across years).

Aggregation: Class aggregates Students via StudentEnrollmentHistory (enrollment is the source of truth; Student can exist without Class).

Association: Curriculum links AcademicYear + Grade + Subject (many-to-many bridge).

Planned: CurriculumTemplate, CurriculumTemplateSystem for country-based templates (§11.5).



Relationship summary (Diagram 2):

From	To	Type	Cardinality
School	Grade, Subject, AcademicYear	Composition	1 : 0..*
AcademicYear	Class, Curriculum, StudentEnrollmentHistory	Composition	1 : 0..*
Class	Grade	Association	N : 1
Class	Teacher	Association	N : 0..1 (supervisor)
Curriculum	AcademicYear, Grade, Subject	Association	N : 1 each
StudentEnrollmentHistory	Student, Class, AcademicYear	Association	N : 1 each
Class	Student	Aggregation (via enrollment)	1 : 0..*

Diagram 3: Scheduling & Academic Activities

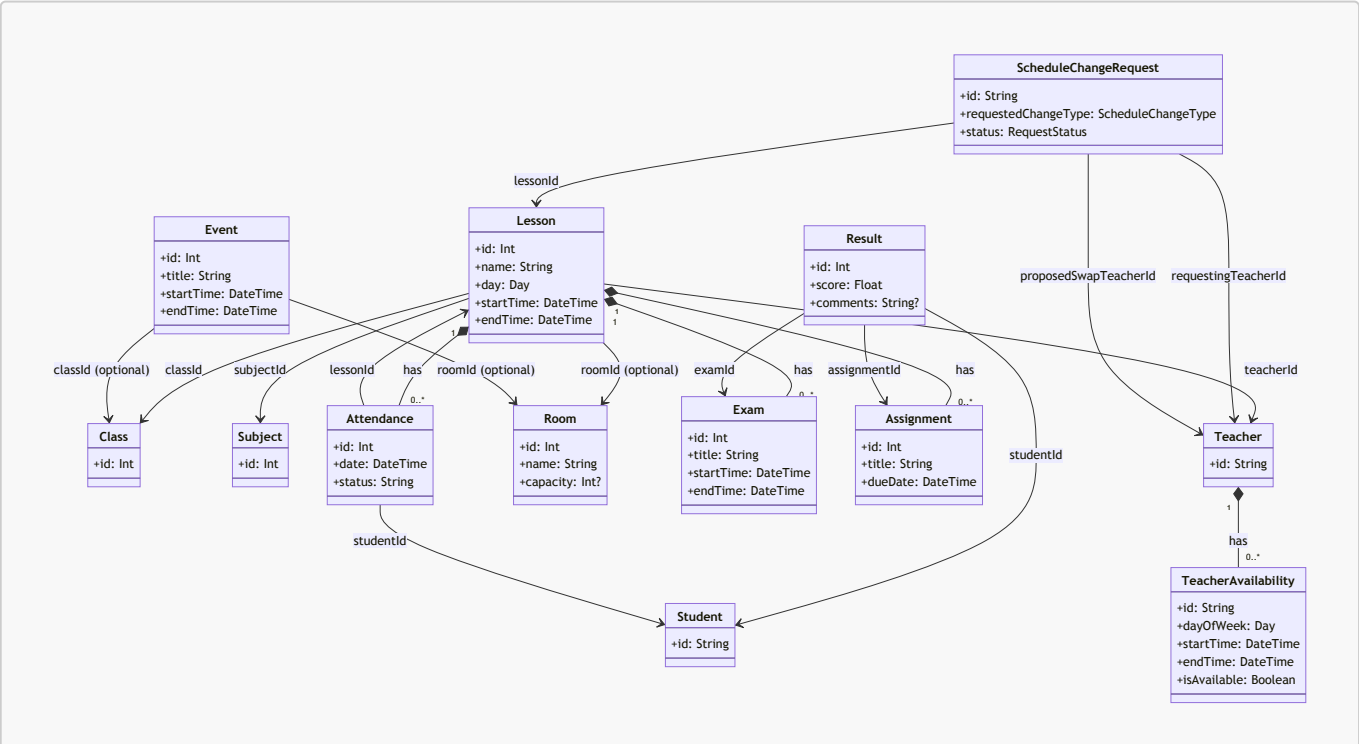
Composition: Lesson *owns* Exam, Assignment, Attendance (they are tied to a lesson and have no meaning without it).

Aggregation: Result aggregates Student with Exam or Assignment (Student and Exam/Assignment can exist independently).

Association: Lesson links Class, Subject, Teacher, Room. Room is optional.

Composition: ScheduleChangeRequest is owned by School and depends on Lesson and Teacher.

Composition: TeacherAvailability is owned by Teacher (availability slots).



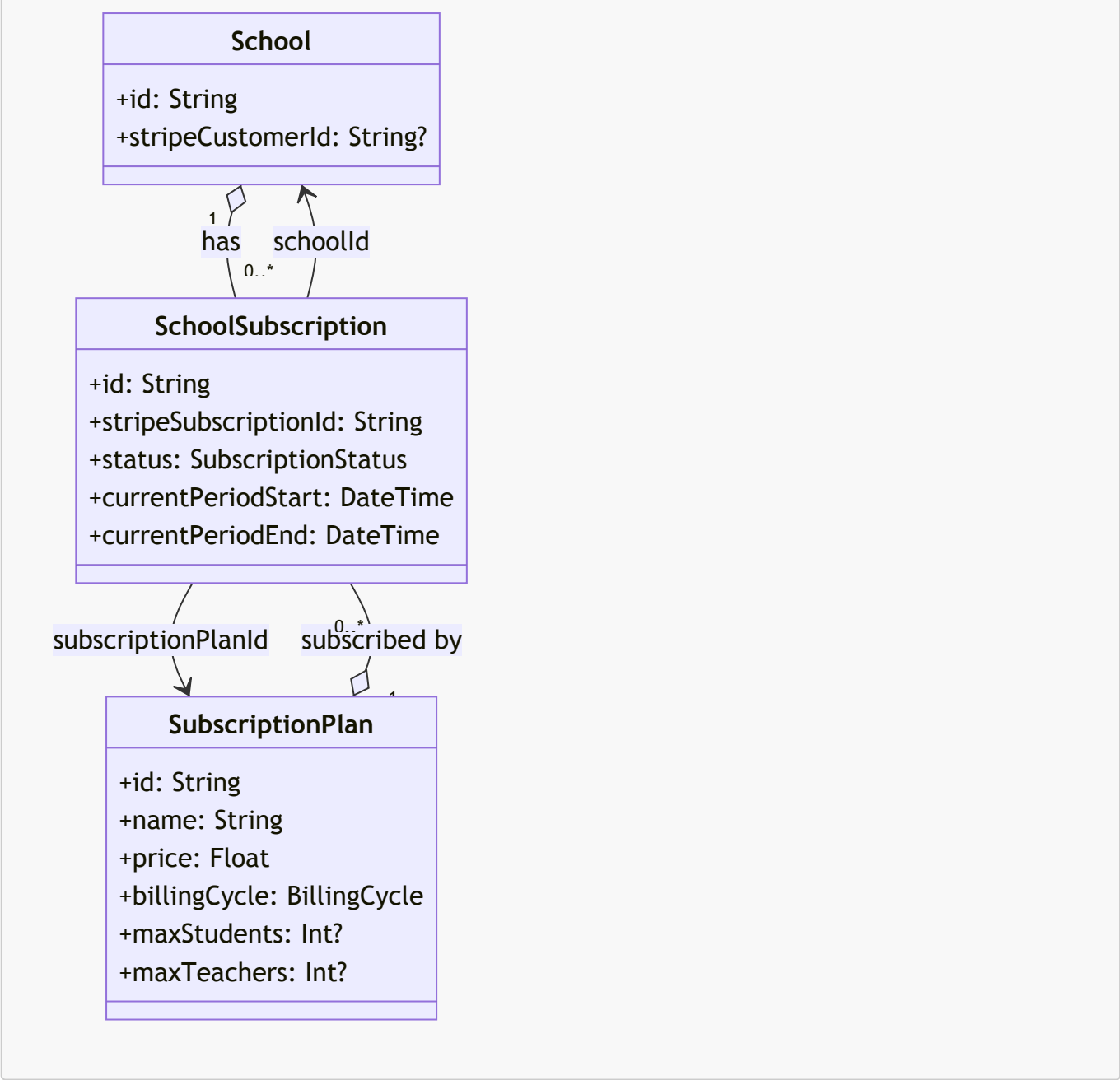
Relationship summary (Diagram 3):

From	To	Type	Cardinality
Lesson	Class, Subject, Teacher	Association	N : 1
Lesson	Room	Association	N : 0..1
Lesson	Exam, Assignment, Attendance	Composition	1 : 0..*
Result	Student, Exam/Assignment	Association	N : 1
Attendance	Student, Lesson	Association	N : 1 each
ScheduleChangeRequest	Lesson, Teacher	Association	N : 1
Teacher	TeacherAvailability	Composition	1 : 0..*

Diagram 4: Billing

Association: SchoolSubscription links School to SubscriptionPlan. School and SubscriptionPlan can exist independently.

Aggregation: SchoolSubscription is the join between School and SubscriptionPlan.



Relationship summary (Diagram 4):

From	To	Type	Cardinality
School	SchoolSubscription	Aggregation	1 : 0..*
SubscriptionPlan	SchoolSubscription	Aggregation	1 : 0..*
SchoolSubscription	School, SubscriptionPlan	Association	N : 1 each

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Diagram 5: Planned – Teacher Marketplace & Student Transfer

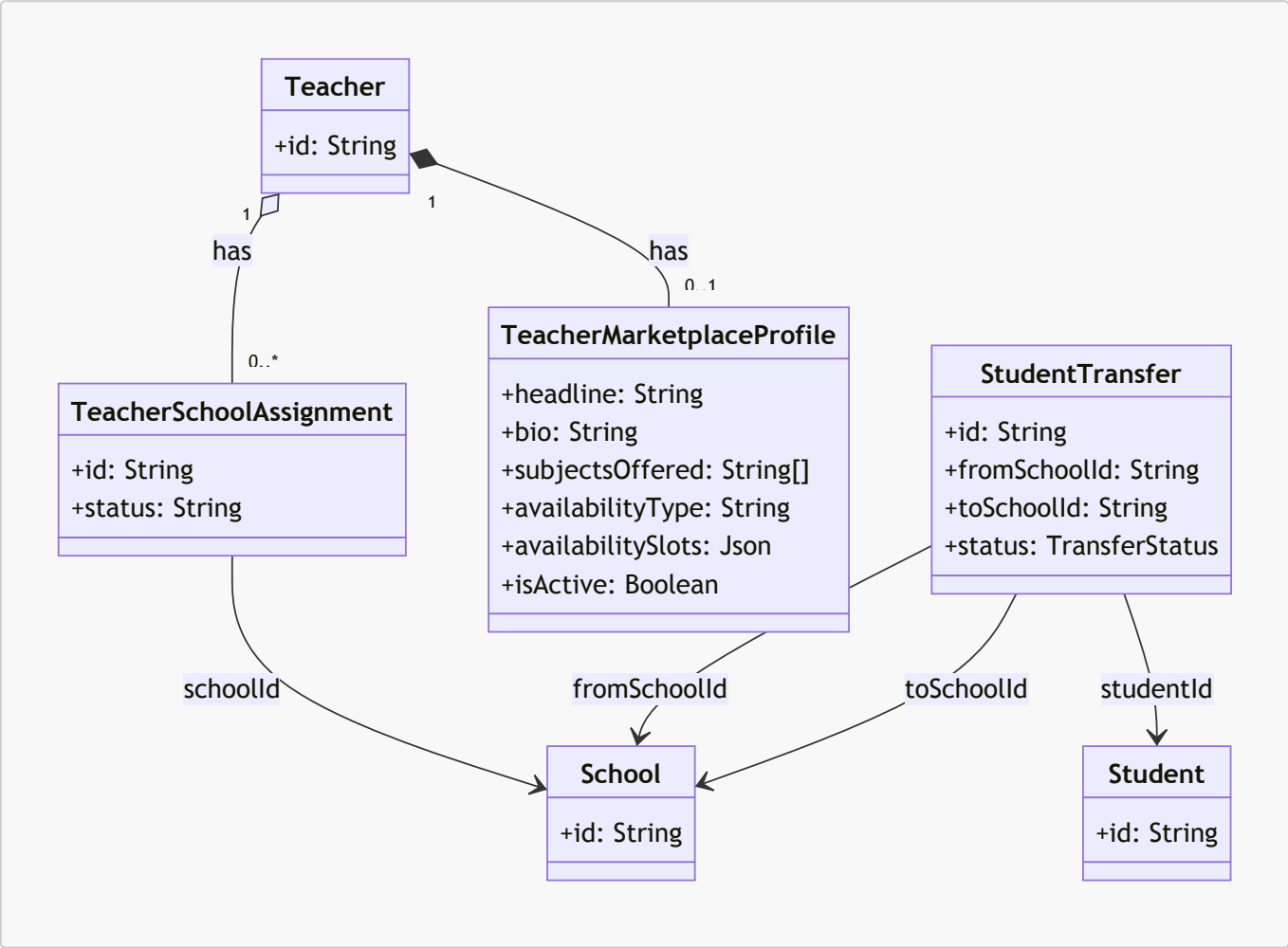
Entities from SYSTEM_DESIGN §14 and §15, not yet in the schema.

Aggregation: Teacher *aggregates* TeacherSchoolAssignment—a teacher can work at multiple schools; each assignment links the teacher to one school. The assignment can exist independently of the teacher's profile (e.g. for historical records).

Composition: Teacher *owns* TeacherMarketplaceProfile—the marketplace profile is an extension of the teacher for discovery; it has no meaning without the teacher.

Association: TeacherSchoolAssignment links to School (the school where the teacher works). StudentTransfer links to Student and to two Schools (fromSchoolId, toSchoolId).

Dependency: StudentTransfer depends on Student and School—when the transfer completes, it updates `Student.schoolId` and `Auth.schoolId`.



Relationship summary (Diagram 5):

From	To	Type	Cardinality
Teacher	TeacherSchoolAssignment	Aggregation	1 : 0..*
Teacher	TeacherMarketplaceProfile	Composition	1 : 0..1
TeacherSchoolAssignment	School	Association	N : 1
StudentTransfer	Student	Association	N : 1
StudentTransfer	School	Association	N : 1 (fromSchoolId)
StudentTransfer	School	Association	N : 1 (toSchoolId)

Cross-Diagram Dependencies

Overview

The five domain diagrams are not isolated—they share entities and form a dependency chain. Understanding these links helps when implementing features, writing queries, or tracing data flow across the system.

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Diagram 1 (Core) ──► Diagram 2 (Academic): School, Auth profiles
Diagram 2 (Academic) ──► Diagram 3 (Scheduling): Class, Subject, Teacher,
Student
Diagram 1 (Core) ──► Diagram 4 (Billing): School
Diagram 5 (Planned) ──► Diagram 1, 2: Teacher, Student, School
```

Diagram 1 → Diagram 2 (Core → Academic)

Shared entities: School, Teacher, Student

Why: The academic structure is scoped to a School. Diagram 2 introduces AcademicYear, Grade, Class, Subject, Curriculum, and StudentEnrollmentHistory—all of which belong to a School. Teacher and Student from Diagram 1 are referenced in Diagram 2:

Diagram 2 entity	Depends on (Diagram 1)	How
AcademicYear	School	`academicYear.schoolId` → School
Grade	School	`grade.schoolId` → School
Subject	School	`subject.schoolId` → School
Class	School, Teacher	`class.schoolId` → School; `class.supervisorId` → Teacher
StudentEnrollmentHistory	Student	`enrollment.studentId` → Student

Implication: You cannot create an AcademicYear, Class, or Curriculum without a School. You cannot enroll a Student in a Class without the Student (and Auth) from Diagram 1.

Diagram 2 → Diagram 3 (Academic → Scheduling)

Shared entities: Class, Subject, Teacher, Student

Why: Scheduling and academic activities build on the academic structure. A Lesson is a time slot for a Class, taught by a Teacher, covering a Subject. Exams, Assignments, Attendance, and Results all depend on these entities.

Diagram 3 entity	Depends on (Diagram 2)	How

Diagram 3 entity	Depends on (Diagram 2)	How
Lesson	Class, Subject, Teacher	`lesson.classId`, `lesson.subjectId`, `lesson.teacherId`
Exam	Lesson	`exam.lessonId` → Lesson (Lesson links to Class, Subject, Teacher)
Assignment	Lesson	`assignment.lessonId` → Lesson
Attendance	Lesson, Student	`attendance.lessonId` → Lesson; `attendance.studentId` → Student
Result	Student, Exam/Assignment	`result.studentId` → Student; `result.examId` or `result.assignmentId`

Implication: You cannot create a Lesson without a Class (which implies AcademicYear), Subject, and Teacher. Attendance and Results require Students who are enrolled in the Lesson's Class (via StudentEnrollmentHistory). The curriculum check (Lesson's Subject must be in the Class's grade curriculum) crosses Diagram 2 and 3.

Diagram 1 → Diagram 4 (Core → Billing)

Shared entities: School

Why: Billing is at the school level. A School subscribes to a SubscriptionPlan via SchoolSubscription. SubscriptionPlan is global (not owned by any School); SchoolSubscription is the join.

Diagram 4 entity	Depends on (Diagram 1)	How
SchoolSubscription	School	`schoolSubscription.schoolId` → School

Implication: A School must exist before it can have a subscription. The School's `stripeCustomerId` (Diagram 1) is used when creating or managing subscriptions.

Diagram 5 → Diagrams 1 & 2 (Planned → Core & Academic)

Shared entities: Teacher, Student, School

Why: The planned Teacher Marketplace and Student Transfer features extend the current model. They introduce new entities but rely on existing ones.

Diagram 5 entity	Depends on	How
TeacherSchoolAssignment	Teacher, School	Links a Teacher to a School; replaces or supplements `Teacher.schoolId` for multi-school support

Diagram 5 entity	Depends on	How
TeacherMarketplaceProfile	Teacher	Extends Teacher with marketplace-specific fields (headline, availabilitySlots, etc.)
StudentTransfer	Student, School	`fromSchoolId`, `toSchoolId` reference Schools; `studentId` references Student; transfer updates `Student.schoolId`

Implication: TeacherSchoolAssignment changes the Teacher–School relationship from 1:1 to N:N. StudentTransfer changes Student.schoolId when a transfer completes. Parent access may become derived from children's schools (Diagram 1) rather than a single Parent.schoolId.

Dependency Flow (Top-Down)

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                    ┌──────────────┐
                    │  Diagram 1   │
                    │   (Core)     │
                    │ School, Auth│
                    │ Admin, etc.  │
                    └──────────────┘
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          ┌────────────────┼────────────────┐
          │                │                │
          ▼                ▼                ▼
   ┌──────────┐      ┌──────────┐      ┌──────────┐
   │Diagram 2 │      │Diagram 4 │      │Diagram 5 │
   │(Academic)│      │(Billing) │      │(Planned) │
   └──────────┘      └──────────┘      └──────────┘
        │
        ▼
   ┌──────────┐
   │Diagram 3 │
   │(Schedul.)│
   └──────────┘
```

Order of creation (typical): School → Auth + profiles → AcademicYear → Grade, Subject → Class → Curriculum → StudentEnrollmentHistory → Lesson → Exam, Assignment, Attendance, Result.

Reverse Dependencies (Queries & Navigation)

Dependencies also flow in reverse when reading data:

- **"Show a student's schedule"**: Student (1) → StudentEnrollmentHistory (2) → Class (2) → Lesson (3)
- **"List lessons for a teacher"**: Teacher (1) → Lesson (3) → Class, Subject (2)

- **"Check if school can add more students"**: School (1) → SchoolSubscription (4) → SubscriptionPlan (4) → maxStudents